MTH631 - Quiz No.4

Date: 09 January 2026

Q.1 \begin{gathered} {\text{In }}{\mathbb{R}^2},f\left({x,y} \right){\text{ = }}\left\{ \begin{gathered} \frac{{\sin \sqrt {1 - {x^2} - 2{y^2}} }}{{\sqrt {1 - {x^2} - 2{y^2}} }},\,\,{x^2} + 2{y^2} < 1 \hfill \\ 1,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,{x^2} + 2{y^2} = 1 \hfill \\\ \end{gathered} \right. \hfill \\ {\text{is - - - - - - - - - - on }}{x^2} + 2{y^2} = 1{\text{ }}{\text{.}} \hfill \\\ \end{gathered}

A. continuous

B. discontinuous

Q.2 {\text{The function }}d{x_i}{\text{ gives the value at a point in }}{\mathbb{R}^n}{\text{as;}}

A. d{x_i}(X) = {x_i}

B. d{x_i}(X) = {x_{i - 1}}

C. d{x_i}(X) = {x_{i - 1}} + {x_{i + 1}}

D. d{x_i}(X) = {x_{i + 1}}

Q.3 \begin{array}{*{20}{l}} {{\text{Let }}f{\text{ be defined on an interval }}I{\text{ in }}{\mathbb{R}^2}.\,{\text{Suppose that }}{x_1}{\text{ and }}{x_{\text{2}}}{\text{ are in }}I{\text{ and}}\,f\left({{x_1}} \right){\text{ }} < {\text{ }}y{\text{ }} < {\text{ }}f\left({{x_2}} \right){\text{. }}} \\\ {{\text{Then }}f\left(x \right) = y{\text{ for some }}x{\text{ in }}I.} \\\ {} \end{array}

A. True

B. False

Q.4 Identify the false statement(s)

A. None of these

B. For a functions of n variables f({{X}}) the partial derivative with respect to variable {x_2} is {{ f({{X}})}}{{ {{E}}_2}}}

C. The partial derivative of the function f(x,y,z) = 3xyz + 2{x^2} + {z^2} with respect to third variable is {f_z} = 2z

D. The partial derivative is a special case of directional derivative.

Q.5 \begin{gathered} {\text{Analogous to the derivative of a function of one variable in }}{\mathbb{R}^2}{\text{, the directional derivative of a function}} \hfill \\ f{\text{ at }}{X_0}{\text{ in }}{\mathbb{R}^n}{\text{ has - - - - - - value(s)}}{\text{. }} \hfill \\\ \end{gathered}

A. integral

B. multiple

C. infinite many

D. unique

Q.6 \begin{gathered} {\text{Let }}u{\text{ and }}v{\text{ be functions of two variables with continuous second - order partial derivatives in a region }}S{\text{. }} \hfill \\ {\text{Suppose that }}{u_x} = {v_y}{\text{ and }}{u_y} = - {v_x}{\text{ in }}S{\text{. Then, }}{v_{xx}} + {v_{yy}} - - - - . \hfill \\ \hfill \\\ \end{gathered}

A. =1

B. >0

C. =0

D. >0 but <1

Q.7 For the functions f(x) = 3{x^2} + 5{x^3}, the differential at x = - 1 exists because

A. { }_{x - 1} {{f(x) + 2 - 9(x + 1)}}{{x + 1}} = 1.

B. { }_{x - 1} {{f(x)}}{{x + 1}} = .

C. { }_{x - 1} {{f(x) + 2 - 9(x + 1)}}{{x + 1}} = 0.

D. { }_{x - 1} {{9(x + 1)}}{{x + 1}} = 0.

Q.8 {\text{If }}f\left(t \right) = \sqrt t {\text{, }}g\left({x,y} \right) = 1 - {x^2} - 2{y^2}{\text{, then the domain of }}f \circ g = {\text{ - - - - - - }}{\text{.}}

A. \{ {({x,y}):{x^2} - 2{y^2} 1} \}

B. {R}

C. {{R}^2}

D. \{ {({x,y}):{x^2} - 2{y^2} 1} \}

Q.9 Let the function f(s) = \left\{ \begin{array}{l}\frac{{\sin s}}{s},s \ne 0\\1,s = 0\end{array} \right. is continuous over domain ____________.

A. (a) {R}

B. Both a & b

C. (c) (0,)

D. (b) (- ,)

Q.10 \begin{gathered} {\text{The domain of the real valued function }}f:{\mathbb{R}^n} \to \mathbb{R}{\text{ defined and given by;}} \hfill \\ f\left(X \right){\text{ = }}{\left({1 - x_1^2 - x_2^2 - \cdots - x_2^n} \right)^{ - 1}}{\text{ is - - - - - }}{\text{.}} \hfill \\\ \end{gathered}

A. \{ {X| | X | = 1} \}

B. \{ {X| | X | 1} \}
